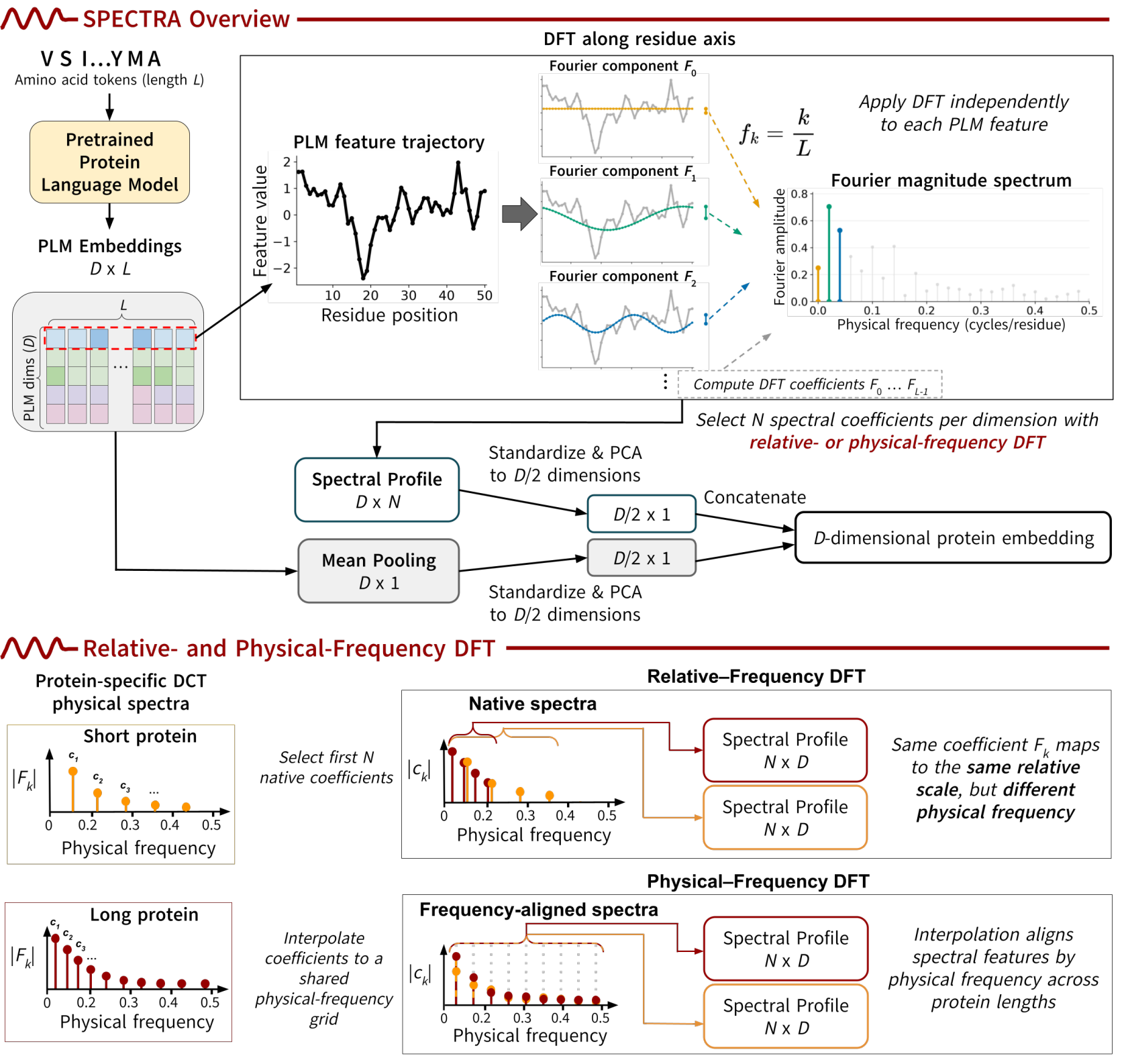

SPECTRA Overview
DFT along residue axis
V S I...Y M A
Amino acid tokens (length L)
Pretrained Protein Language Model
PLM Embeddings
D x L
L
PLM dims (D)
PLM feature trajectory
Feature value
Residue position
Fourier component F0
Fourier component F1
Fourier component F2
$f_k = \frac{k}{L}$
Apply DFT independently to each PLM feature
Fourier magnitude spectrum
Fourier amplitude
Physical frequency (cycles/residue)
Compute DFT coefficients F0 ... FL-1
Select N spectral coefficients per dimension with relative- or physical-frequency DFT
Spectral Profile
D x N
Standardize & PCA to D/2 dimensions
D/2 x 1
Concatenate
D-dimensional protein embedding
Mean Pooling
D x 1
D/2 x 1
Standardize & PCA to D/2 dimensions
Relative- and Physical-Frequency DFT
Protein-specific DCT physical spectra
Short protein
|Fk|
Physical frequency
Select first N native coefficients
Relative–Frequency DFT
Native spectra
|ck|
Physical frequency
Spectral Profile N x D
Spectral Profile N x D
Same coefficient Fk maps to the same relative scale, but different physical frequency
Long protein
Physical–Frequency DFT
Frequency-aligned spectra
Interpolate coefficients to a shared physical-frequency grid
Interpolation aligns spectral features by physical frequency across protein lengths